X5D: A Polyhedral Reinterpretation of the Exponent Database as a Five-Dimensional Geometric Flow

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Abstract

We reinterpret the Analytic Number Theory Exponent Database (EXPDB) of Tao, Trudgian, and Yang as a monotone contraction flow on a single five-dimensional bounded rational polyhedral region $\mathcal{P} \subset \mathbb{R}^5$ in coordinates $(\sigma, \tau, \rho, \rho^*, s)$. The principal downstream objects computed by the EXPDB pipeline — the large value region, the zeta large value region, the energy region, the zero density estimate $A(\sigma)$, the zero density energy estimate $A^*(\sigma)$, and the prime gap bound θ — are obtained from $\mathcal{P}$ by projections, rational suprema, and envelopes. The upstream two-dimensional objects (the exponent pair hull, the beta envelope, and the mu hull) generate constraints *on* $\mathcal{P}$ rather than descending from it. We formalize the collection of boundary objects as the *X5D signature*, prove it is a monotone invariant of a contraction flow on a product lattice, and establish finite convergence to a unique fixed point via a descending chain condition argument. An appendix maps each code module to its geometric role.

1 Introduction

1.1 The Exponent Database

The Analytic Number Theory Exponent Database (EXPDB) is a computational system, initiated by Tao, Trudgian, and Yang [TTY25], that mechanizes the derivation of bounds on exponents arising throughout analytic number theory. It records, in both human-readable and executable form, a corpus of literature results — exponent pairs, bounds on exponential sum growth rates, large value estimates, zero density estimates, additive energy estimates, and others — together with the network of transformations that relate them. Given an input set of axioms, EXPDB applies a cascade of transformations and produces the tightest bounds derivable from those axioms.

The system is implemented as approximately 10,700 lines of Python organized into 24 modules. Its core data structure is the *Hypothesis* object: a typed mathematical assertion carrying a data payload, a human-readable proof, a bibliographic reference, and a dependency set. The system supports ten hypothesis types (upper bound on beta, exponent pair, exponent pair transform, upper bound on mu, large value estimate, zeta large value estimate, zero density estimate, zero density energy estimate, large value energy region, large value energy region transform).

The architecture of EXPDB is procedural: derivation routines call transformation functions in a prescribed sequence, accumulating new hypotheses in a mutable set. The present paper proposes an alternative perspective. We argue that the entire pipeline operates, at every stage, on projections and envelopes of a single geometric object — a five-dimensional polyhedral region — and that this reinterpretation clarifies the system's convergence behavior, its information-theoretic structure, and its points of contact with other geometric frameworks.

Terminology: X5D, EXPDB, and ANTEDB. We distinguish three names. *EXPDB* refers to the Analytic Number Theory Exponent Database — the repository and analytic

pipeline described above. *ANTEDB* refers to the broader Analytic Number Theory project umbrella under which EXPDB and related computational tools are organized. *X5D* refers to the cross-framework geometric method — the five-dimensional polyhedral framework developed in this paper — that spans EXPDB and ANTEDB. When we speak of the "X5D framework" or an "X5D signature," we mean the geometric machinery of Sections 4–7, not the EXPDB codebase itself.

1.2 The Geometric Thesis

The central claim of this paper is:

> **Thesis.** There exists a bounded rational polyhedral region $\mathcal{P} \subset \mathbb{R}^5$, representable as a finite disjoint union of convex polytopes cut out by rational half-spaces, such that the principal *downstream* objects computed by the EXPDB pipeline — the large value region, the zeta large value region, the energy region, the zero density estimate, the zero density energy estimate, and the prime gap bound — arise as projections, rational suprema, or envelopes of $\mathcal{P}$. The *upstream* two-dimensional objects (the exponent pair hull, the beta envelope, and the mu hull) are not descended from $\mathcal{P}$; they are generators of constraints *on* $\mathcal{P}$. A small number of direct EP-based routes to the zero density estimate also bypass $\mathcal{P}$ and contribute additional rational curves to the final lower envelope.

The five coordinates of the ambient space are $(\sigma, \tau, \rho, \rho^*, s)$, corresponding to the variables that parameterize the *large value energy region* (LVER), the highest-dimensional object in the EXPDB type system. Concretely, $\mathcal{P}$ is the feasible set of all five-tuples $(\sigma, \tau, \rho, \rho^*, s)$ that are simultaneously consistent with every constraint derivable from the input hypotheses.

Every *downstream* bound that EXPDB produces can be written as a composition of three operations: (i) *projection*, dropping coordinates to obtain a lower-dimensional shadow; (ii) *rational supremum*, computing $\sup_\tau\{\rho/\tau\}$ over a projected region; and (iii) *lower envelope*, taking the pointwise minimum over a family of such functions. The full downstream derivation chain descends the *dimension ladder*

$$\mathcal{P} \subset \mathbb{R}^5 \xrightarrow{\pi} R \subset \mathbb{R}^3 \xrightarrow{\sup \rho/\tau} f(\sigma) \xrightarrow{\inf} f^*(\sigma) \xrightarrow{\sup_\sigma} \theta \in \mathbb{R}.$$

The 2D stage does not appear in this descending chain: the EP hull, beta envelope, and mu hull sit in $\mathbb{R}^2$ but flow *upward* into $\mathcal{P}$ as constraint generators, not downward from it. We call the collection of boundary functions obtained at each stage the *X5D signature*, and we show that it is a monotone invariant of a contraction flow on a product lattice.

1.3 Why a Geometric Reinterpretation?

The procedural description of EXPDB leaves several structural questions unanswered. **Convergence:** the pipeline contains two feedback loops (EP $\leftrightarrow$ beta, and LVER $\to$ LV $\to$ LVER); why do they terminate and what is the limiting object? The polyhedral framework answers this directly: the loops correspond to alternating projections and liftings of $\mathcal{P}$, and they converge because $\mathcal{P}$ is a fixed point of the composition. **Information loss:** how much information is lost at each stage? Each projection discards constraints involving the dropped coordinates, and the loss is irreversible. **Duality:** the EP hull and the beta envelope are related by a classical projective duality; the X5D framework embeds this duality into a section-projection adjunction between $\mathbb{R}^2$ and $\mathbb{R}^5$. **Optimality:** the bounds are optimal *among those derivable from the available half-spaces defining* $\mathcal{P}$; improving them requires adding new half-spaces or changing the coordinate system.

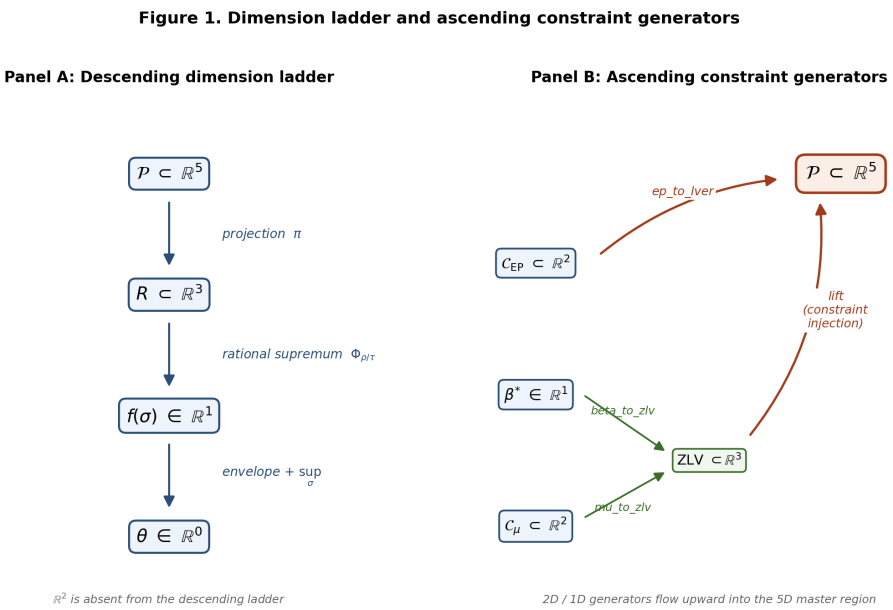

Figure 1: Dimension ladder and ascending constraint generators.

1.4 Scope and Limitations

We *do* provide a complete structural analysis of the EXPDB codebase, define the master polyhedral region $\mathcal{P}$, formalize the X5D signature, and prove its convergence. We do *not* engage with the underlying mathematics of analytic number theory. All references to coordinates $(\sigma, \tau, \rho, \rho^*, s)$ should be understood as abstract parameters of a five-dimensional ambient space. We do *not* propose modifications to EXPDB; our reinterpretation is descriptive.

1.5 Relation to Prior Work

The EXPDB system itself is described in [TTY25]. The computational infrastructure relies on `pycddlib` [F] for exact rational polytope arithmetic and `sympy` [MSP+17] for symbolic manipulation of rational functions. The idea of reinterpreting a derivation system as geometry on a polyhedral object has precedents in linear programming (simplex method), tropical geometry, and the theory of Newton polytopes.

1.6 Outline

Section 2 fixes notation and collects definitions from polyhedral geometry, piecewise-function theory, and order-theoretic lattices. **Section 3** describes the EXPDB pipeline as a computational system. **Section 4** defines the master polyhedral region $\mathcal{P}$ and establishes its properties. **Section 5** shows that every downstream EXPDB object is a projection, section, or envelope of $\mathcal{P}$. **Section 6** defines the X5D signature formally, proves it is a monotone invariant, and establishes convergence. **Section 7** develops consequences: the dimension ladder, the duality web, the flow structure, and the fixed-point hierarchy. **Section 8** discusses future directions. **Appendix A** provides a complete mapping from code modules to geometric operations.

2 Preliminaries: Polyhedral Geometry over the Rationals

All objects are defined over the rational field $\mathbb{Q}$, reflecting the exact-arithmetic design of EXPDB.

2.1 Polytopes

Definition 2.1 (Half-space)**.** *A* rational half-space *in* $\mathbb{R}^d$ *is* $H^+ = \{x \in \mathbb{R}^d : a_0 + a_1x_1 + \cdots + a_dx_d \geq 0\}$ *with* $a_i \in \mathbb{Q}$*.*

Definition 2.2 (H-polytope)**.** *An* H-polytope $P \subset \mathbb{R}^d$ *is a finite intersection of rational half-spaces.*

Definition 2.3 (V-polytope)**.** *A* V-polytope *is the convex hull of a finite set of rational points.*

The double description method [MRTT53, F] converts between the H- and V-representations. All polytopes in this paper are bounded, guaranteed by a global bounding box imposed by truncation constants.

2.2 Operations on Polytopes

We record the operations used: **intersection** (the meet in the inclusion lattice of convex bodies), **set difference** (decomposed into a disjoint union of polytopes), **scaling** (coordinatewise rational rescaling), **projection** (image under a coordinate-erasing linear map; preserves convexity), **lifting** (Cartesian product with additional coordinate intervals; right adjoint to projection), and **containment/emptiness testing** via LP.

2.3 Regions

Many feasible sets are not convex. The *region* data structure represents arbitrary boolean combinations of polytopes (atoms, complements, unions, disjoint unions, intersections). Every region can be canonicalized as a disjoint union of convex polytopes via the distributive law.

2.4 Affine and Piecewise-Affine Functions

Definition 2.4 (Piecewise-affine function)**.** *A finite collection* $\{(a^{(j)}, P_j)\}$ *of affine functions on polytope domains with pairwise disjoint interiors, covering the full domain.*

The *lower envelope* $f^*(x) = \min_j f_j(x)$ of n affine functions on $\mathbb{R}$ is a concave piecewise-affine function with at most $n-1$ breakpoints; the *upper envelope* is dually convex.

2.5 Rational Functions and Their Envelopes

A *rational function* over $\mathbb{Q}$ is a quotient $r(x) = p(x)/q(x)$. A *piecewise-rational function* is a finite collection $\{(r_j, I_j)\}$ on an interval partition. The *lower envelope* has breakpoints at the real roots of $p_i(x)q_j(x) - p_j(x)q_i(x)$ and is computable by `sympy.solve`.

Remark 2.5 (Origin of rational functions)**.** *Rational functions arise in EXPDB from the* supremum of a ratio *operation. Along each polytope edge,* $\rho(\sigma)$ *and* $\tau(\sigma)$ *are affine in* σ*, so their ratio is a degree-*$(1,1)$ *rational function.*

2.6 The Rational Supremum Operator

Definition 2.6 (Rational supremum)**.** *Let* $R \subset \mathbb{R}^3$ *be a bounded region in coordinates* (σ, τ, ρ)*. The* rational supremum of ρ/τ over R *is the function*

$$\Phi(\sigma) = \sup\{\rho/\tau : (\sigma, \tau, \rho) \in R,\ \tau > 0\}.$$

Proposition 2.7. $\Phi(\sigma)$ *is a piecewise-rational function of* σ *with rational breakpoints.*

Proof sketch. The supremum of a linear-fractional function over a convex polytope is attained along an edge. Along each edge, τ and ρ are affine functions of σ, so their ratio is a degree-$(1,1)$ rational function. The supremum over finitely many such functions is piecewise-rational. □

2.7 Convex Hulls

The convex hull of a finite rational point set is a polytope whose vertices are points that are extreme. In two dimensions, the convex hull of a point set is *dual* to the lower envelope of a family of lines via the Legendre–Fenchel transform: given points (k_i, l_i), the lines $\beta_i(\alpha) = (l_i - k_i)\alpha + k_i$ have lower envelope whose breakpoints biject with hull edges and whose affine pieces biject with hull vertices. This duality is the geometric basis of the EP $\leftrightarrow$ Beta correspondence.

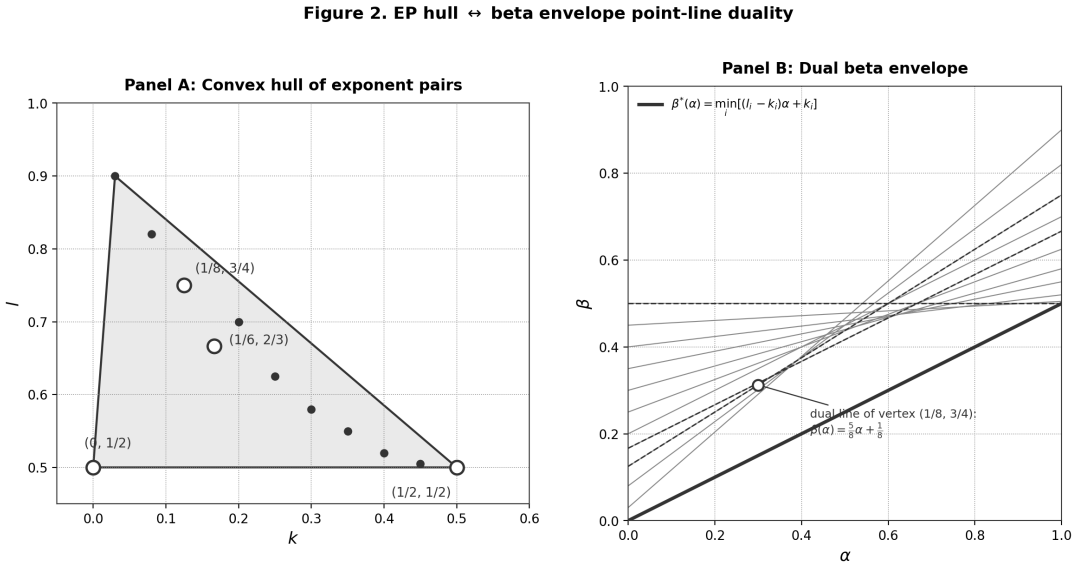

Figure 2: EP hull $\leftrightarrow$ beta envelope point–line duality.

2.8 Order-Theoretic Framework

We equip the space of bounded rational regions, piecewise-rational functions, and scalar bounds with their natural partial orders, where "tighter" corresponds to smaller (for regions and functions) or larger (for 2D hulls). The *EXPDB state lattice* $\mathcal{L}$ is the product of these component lattices, one per X5D component.

Definition 2.8 (Monotone contraction)**.** *A map* $\Phi : \mathcal{L} \to \mathcal{L}$ *is* monotone *if it preserves the product order, and a* contraction *if* $\Phi(\Omega) \preceq \Omega$ *for all reachable states. A monotone contraction on a lattice with descending chain condition converges in finitely many steps.*

3 The EXPDB Pipeline

3.1 The Hypothesis Object

The atomic unit of the pipeline is the *Hypothesis* (`hypotheses.py`). It carries a name, a type tag, a domain-specific payload, a proof string, and a `Reference`. Each hypothesis also carries a mutable `dependencies` set, and the transitive closure of this relation forms a DAG rooted at the derived hypothesis, constituting a machine-readable proof.

3.2 The Type System

Each Hypothesis carries a `hypothesis_type` string drawn from ten type tags: `Upper bound on beta` (1D affine), `Exponent pair` (point in $\mathbb{R}^2$), `Exponent pair transform` (higher-order map), `Exponent pair to beta bound transform`, `Upper bound on mu` (point in $\mathbb{R}^2$), `Large`

`value estimate` (region in $\mathbb{R}^3$), `Zeta large value estimate`, `Large value energy region` (region in $\mathbb{R}^5$), `Zero density estimate` (rational function on interval), and `Zero density energy estimate`.

3.3 The Hypothesis Set

A `Hypothesis_Set` is a mutable collection of Hypothesis objects with insertion, filtering, search, and caching interfaces. It serves as the pipeline's working memory.

3.4 The Axiom Layer: `literature.py`

The module `literature.py` (1,439 lines) populates a module-level `Hypothesis_Set` called `literature` at import time. It proceeds through the hypothesis types in order (beta bounds, exponent pairs, EP transforms, mu bounds, LV estimates, LVER constraints, ZD estimates, ZDE estimates). Approximately 15–20 direct LVER hypotheses arise from the work of Heath-Brown [7], Ivić [8], and Guth–Maynard [10].

3.5 The Transformation Catalog

The transformations are grouped by stage: *within the EP stage* (A-, B-, C-processes and iterative expansion); *EP → Beta* (tangent-line generation and the D-process); *within the Beta stage* (VdC-β iteration and lower-envelope computation); *Beta → EP* (hull-edge extraction); *EP/Beta → Mu* (linear map and functional equation); *into the 3D stage* (LV intersection, raise-to-power, Bourgain optimization, LV/Beta/Mu → ZLV); *into the 5D stage* (EP → LVER as 9 halfspace constraints, LV → LVER as Cartesian lift, LVER raise-to-power, LVER intersection); and *descending 5D → 3D → 1D → 0D* (projections, rational suprema, envelopes, scalar extraction).

3.6 Data Representations

Each hypothesis type has a specific representation. Exponent pairs and mu bounds are points in $\mathbb{Q}^2$; beta bounds are affine functions on intervals; LV, ZLV, and LVER are `Region` objects in $\mathbb{Q}^3$ or $\mathbb{Q}^5$; ZD and ZDE estimates are piecewise-rational functions stored with lazy `sympy` evaluation.

3.7 The Orchestration Layer

The pipeline is driven by `derived.py` (1,384 lines). The top-level entry point is `derived.py:prove_all()`, whose call graph activates the EP, LV, ZD, ZDE, and θ stages in order. Each `prove_*` routine follows a common pattern: select axioms, expand via transforms, optimize via a `best_*` function, and report.

3.8 Truncation and Boundedness

Six constants in `constants.py` serve as global termination guarantees and geometric bounding parameters: `BETA_TRUNCATION`, `EXP_PAIR_TRUNCATION`, `LARGE_VALUES_TRUNCATION` (each = 20), `ALPHA_UPPER_LIMIT` (= 100), `TAU_UPPER_LIMIT` (= 10^6), and `LV_DEFAULT_UPPER_BOUND` (= 10^6). The last three define the bounding box $\mathcal{B}$ within which the master polyhedral region $\mathcal{P}$ is constructed.

3.9 Summary: The Pipeline as Geometry

The axiom layer populates the pipeline with points in $\mathbb{R}^2$, affine segments, and polyhedral regions in $\mathbb{R}^3$ and $\mathbb{R}^5$. The transformation layer applies three families of operations: **hull and envelope**, **intersection**, and **projection and lifting**. The orchestration layer sequences these

operations. The claim of the next section is that this entire sequence can be understood as the progressive construction and then progressive dissection of a single polyhedral region $\mathcal{P} \subset \mathbb{R}^5$.

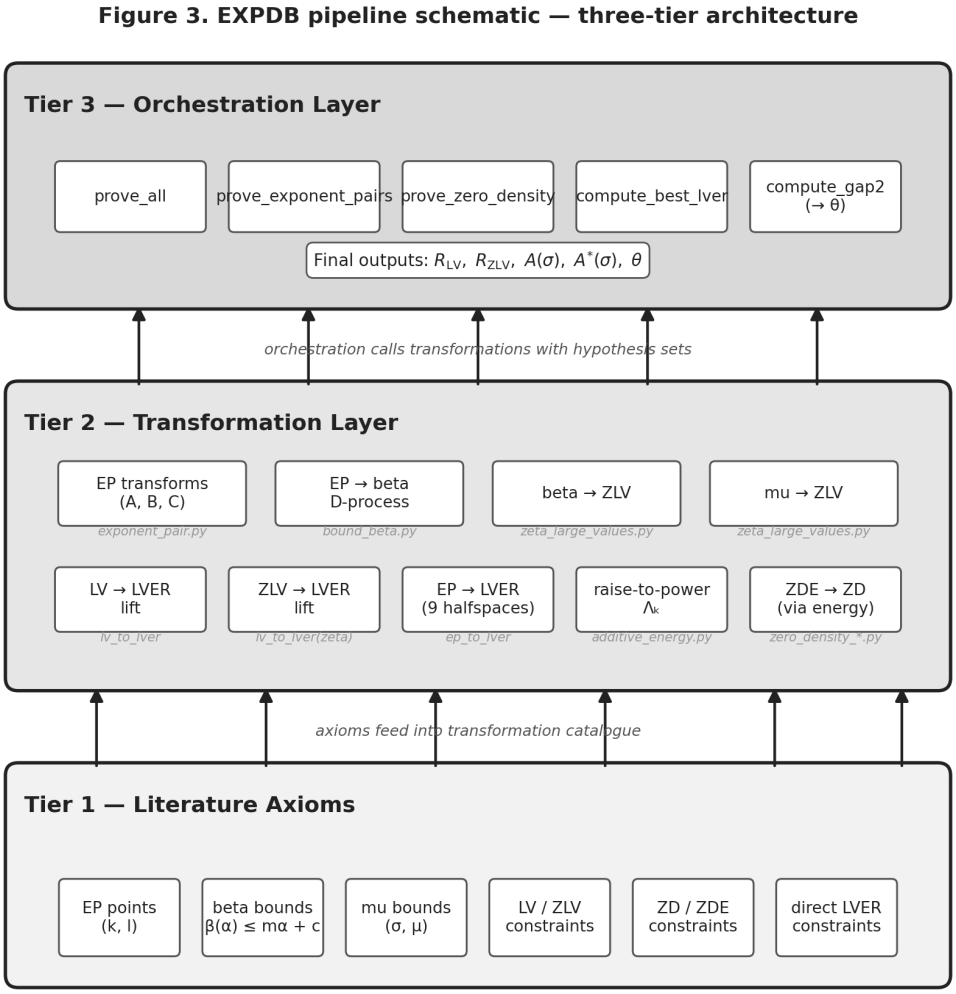

Figure 3: Three-tier pipeline schematic.

4 The Master Polyhedral Region $\mathcal{P} \subset \mathbb{R}^5$

4.1 The Ambient Space

The five-dimensional ambient space is coordinated by $(\sigma, \tau, \rho, \rho^*, s)$: σ is the primary parameter; τ is a scale parameter; ρ projects to the LV region; ρ^* projects to the energy region; and s is an auxiliary parameter eliminated by projection in every output stage.

Remark 4.1 (Why five dimensions)**.** *The coordinate set $(\sigma, \tau, \rho, \rho^*, s)$ is not an arbitrary choice: it is the* minimal *set needed to express the joint constraints currently available in the literature. The triples (σ, τ, ρ) and (σ, τ, ρ^*) are each required to express large value estimates and energy estimates respectively. These two triples could be combined in $\mathbb{R}^4$, but the direct LVER constraints from Heath-Brown [7], Ivić [8], and Guth–Maynard [10] impose joint relations of the form $s \leq f(\rho, \tau)$ and $\rho^* \leq g(s, \sigma)$ that couple ρ and ρ^* only via an auxiliary variable s. Removing s would sever this coupling. Conversely, no literature result presently in EXPDB introduces a sixth variable not expressible in terms of $(\sigma, \tau, \rho, \rho^*, s)$.*

4.2 The Bounding Box

The bounding box

$$\mathcal{B} = [\tfrac{1}{2}, 1] \times [0, T] \times [0, M]^3 \ \subset \ \mathbb{R}^5,$$

with $T = \texttt{TAU_UPPER_LIMIT} = 10^6$ and $M = \texttt{LV_DEFAULT_UPPER_BOUND} = 10^6$, serves as the initial feasible region.

4.3 Sources of Constraints

The master region $\mathcal{P}$ is obtained by intersecting $\mathcal{B}$ with half-spaces from four sources.

4.3.1 Direct LVER constraints (literature).

Several literature results directly constrain the five-tuple $(\sigma, \tau, \rho, \rho^*, s)$. They are packaged via `Region.from_union_of_halfplanes` as a disjoint union of convex polytopes. Example: the Heath-Brown [7] LVER imposes $s \leq \min(2 + \rho,\ 1 + 2\rho,\ 1 + \frac{1}{2}\tau + \frac{5}{4}\rho)$.

4.3.2 Lifted LV constraints.

Each large value estimate in $\mathbb{R}^3$ is lifted to $\mathbb{R}^5$ by Cartesian product with $[0, M]^2$ in the ρ^* and s coordinates.

4.3.3 Lifted ZLV constraints from Beta and Mu.

Beta bounds and mu bounds first generate constraints in (σ, τ, ρ) space via `beta_to_zlv` and `mu_to_zlv`, then lift to $\mathbb{R}^5$.

4.3.4 EP-derived LVER constraints.

Each exponent pair (k, l) generates 9 half-space constraints in $\mathbb{R}^5$ via `ep_to_lver`, packaged as a disjoint union of up to 9 polytopes.

4.3.5 Raise-to-power constraints.

The raise-to-power transform rescales an existing LVER region by $(1, k, k, k, k)$ for $k \in \{2, 3, 4\}$, creating a rescaled copy that covers a larger range of τ values.

4.4 Construction of $\mathcal{P}$

Notation ($\mathcal{P}$-variants). We collect the variants used in the paper. (i) $\mathcal{P} = \mathcal{P}(\mathcal{H}, \mathcal{D})$ is the master polyhedral region cut out by a hypothesis set $\mathcal{H}$ over a (σ, τ)-domain $\mathcal{D}$. (ii) $\mathcal{P}_\zeta$ is the zeta variant ($\tau \geq 2$ plus zeta-specific LVER). (iii) $\mathcal{P}^{(i)}$ ranges over the finite family generated by varying the domain parameters across calls to `prove_zero_density`. (iv) $\mathcal{P}^*$ is the X5D fixed point. (v) $\mathcal{P}_\infty$ is the conjectural limiting region. (vi) $\mathrm{conv}(\mathcal{P})$ is the single convex polytope obtained as the convex hull of $\mathcal{P}$.

Definition 4.2 (Master polyhedral region)**.** *Let $\mathcal{H}$ be a hypothesis set and $\mathcal{D} \subset \mathbb{R}^2$ a (σ, τ)-domain. Collect the LVER hypotheses, expand by raise-to-power transforms, add lifted LV and ZLV constraints, lift $\mathcal{D}$ to $D \subset \mathbb{R}^5$, and intersect:*

$$\mathcal{P}(\mathcal{H}, \mathcal{D}) \ = \ D \ \cap \bigcap_{R \in \mathcal{L}_2} R.$$

The result is a `Region` *of type* `DISJOINT_UNION`*.*

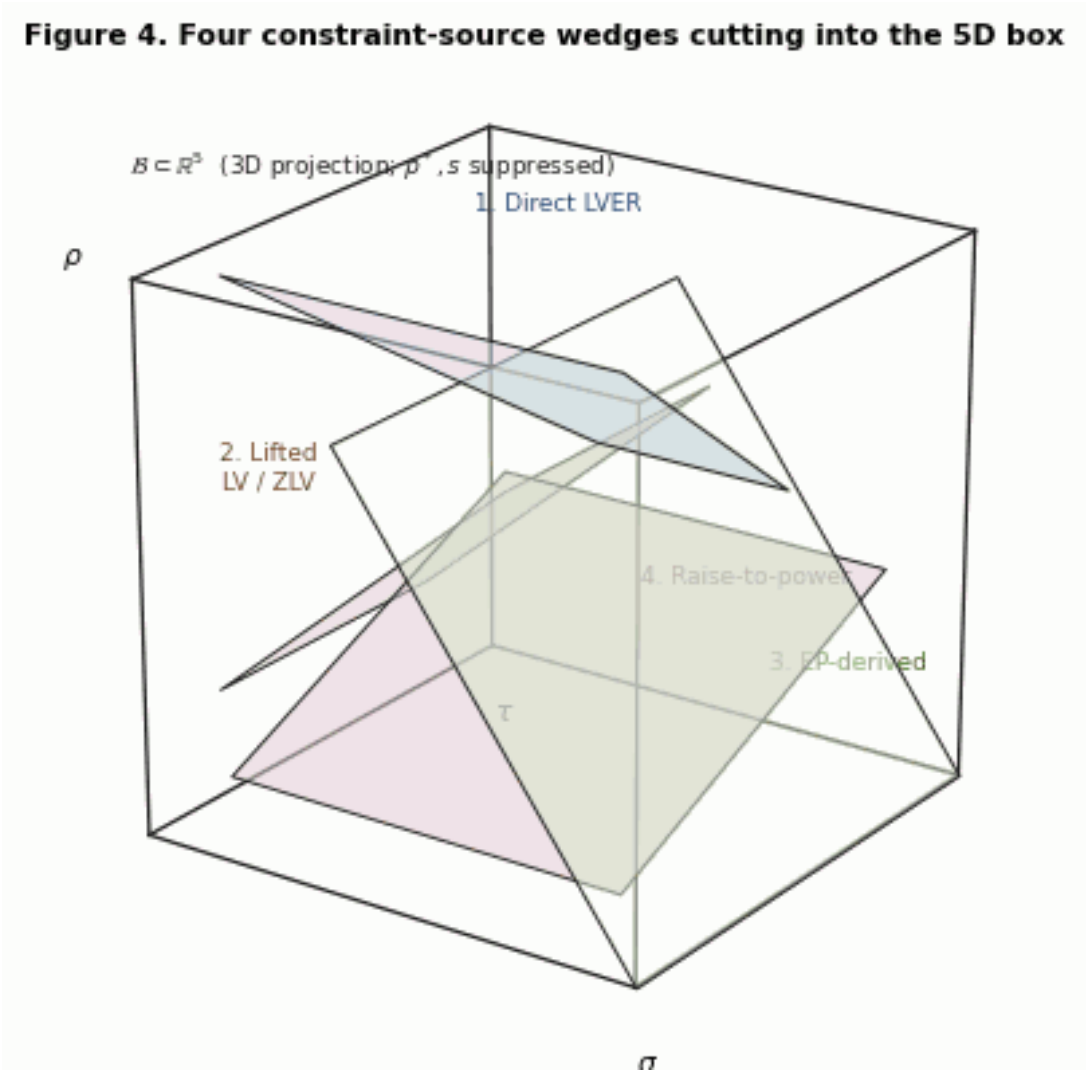

Figure 4: Four constraint-source wedges cutting into the 5D bounding box.

4.5 Properties of $\mathcal{P}$

Proposition 4.3 (Rationality)**.** *Every vertex of every polytope in the disjoint decomposition of $\mathcal{P}$ has rational coordinates.*

Proposition 4.4 (Boundedness)**.** *$\mathcal{P}$ is bounded: $\mathcal{P} \subseteq \mathcal{B}$.*

Proposition 4.5 (Non-convexity of the region; convexity of the pieces)**.** *$\mathcal{P}$ is generally* not *convex as a region. However, its disjoint decomposition $\mathcal{P} = \bigsqcup_j P_j$ consists of convex polytopes P_j.*

Proposition 4.6 (Monotonicity in the hypothesis set)**.** *If $\mathcal{H}_1 \subseteq \mathcal{H}_2$, then $\mathcal{P}(\mathcal{H}_2, \mathcal{D}) \subseteq \mathcal{P}(\mathcal{H}_1, \mathcal{D})$.*

Proposition 4.7 (Dependence on domain)**.** *If $\mathcal{D}_1 \subseteq \mathcal{D}_2$, then $\mathcal{P}(\mathcal{H}, \mathcal{D}_1) \subseteq \mathcal{P}(\mathcal{H}, \mathcal{D}_2)$.*

4.6 Dimension of $\mathcal{P}$

Proposition 4.8. *For any non-trivial hypothesis set and any domain with non-empty interior, $\mathcal{P}$ is generically five-dimensional.*

4.7 The Three Faces of $\mathcal{P}$

The master region has three natural boundary surfaces corresponding to the three types of downstream bounds: the *ρ-boundary* (projects to the LV region and yields $A(\sigma)$), the *ρ^*-boundary* (projects to the energy region and yields $A^*(\sigma)$), and the *s-boundary* (not directly visible in any output but constrains the other two through joint constraints involving s). The interplay between these three boundaries is the source of the pipeline's power: constraints that involve s couple the ρ-boundary and the ρ^*-boundary, so improvements in energy estimates can tighten value estimates and vice versa. This coupling is visible only in the five-dimensional picture.

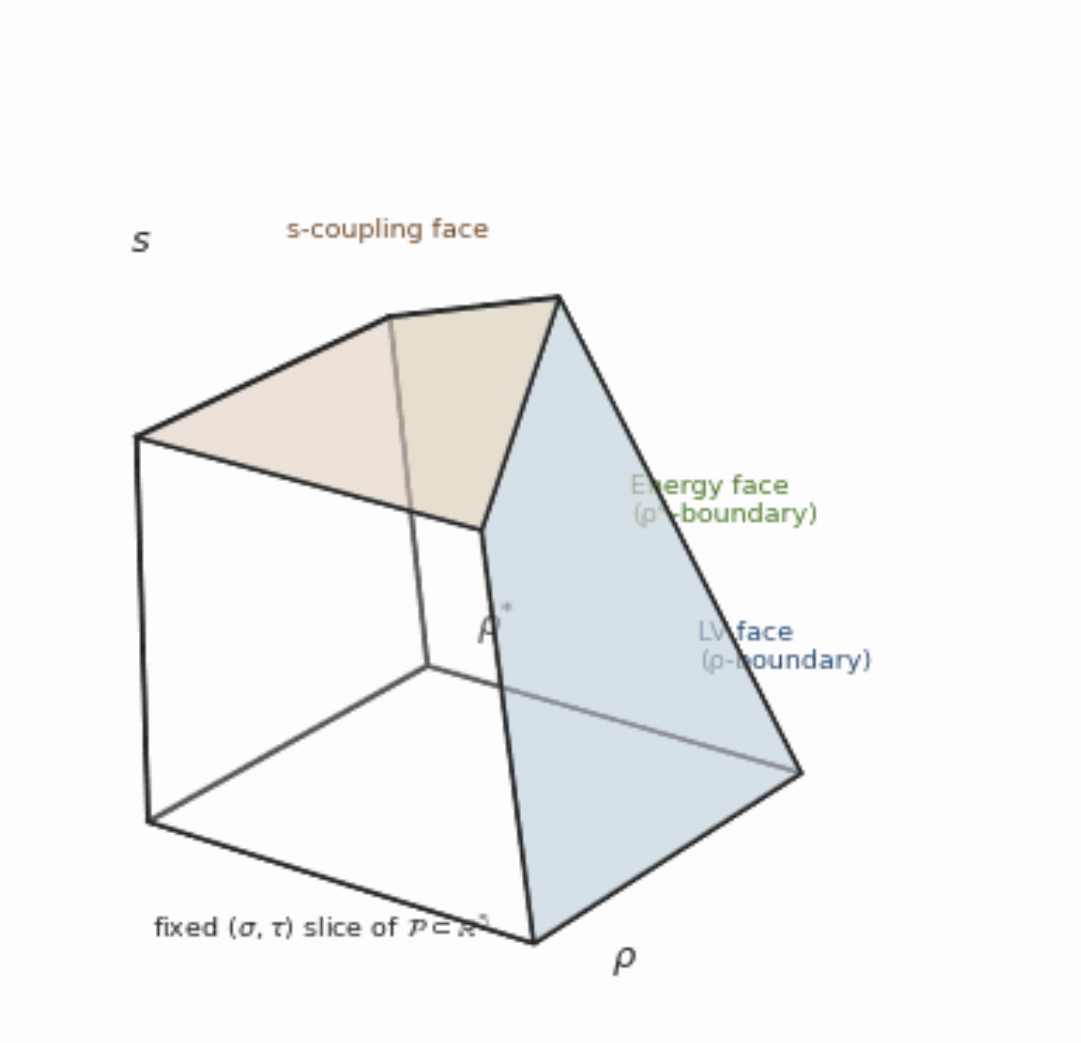

Figure 5: Fixed-(σ, τ) slice of $\mathcal{P}$: LV face, energy face, and s-coupling face.

4.8 $\mathcal{P}$ as the Unique Central Object

Theorem 4.9 (Universality of $\mathcal{P}$)**.** *Every downstream output of the EXPDB pipeline — the LV region, the ZLV region, the energy region, the zero density estimate $A(\sigma)$, the zero density*

energy estimate $A^*(\sigma)$, and the prime gap bound θ — is uniquely determined by $\mathcal{P}(\mathcal{H}, \mathcal{D})$:

$$\begin{aligned}
R_{\mathrm{LV}} &= \pi_{\sigma,\tau,\rho}(\mathcal{P}), \\
R_{\mathrm{energy}} &= \pi_{\sigma,\tau,\rho^*}(\mathcal{P}), \\
A(\sigma) &= \inf_{methods} \sup_{\tau} \{\rho/\tau : (\sigma,\tau,\rho) \in \pi_{\sigma,\tau,\rho}(\mathcal{P})\}, \\
A^*(\sigma) &= \inf_{methods} \frac{1}{1-\sigma} \sup_{\tau} \{\rho^*/\tau : (\sigma,\tau,\rho^*) \in \pi_{\sigma,\tau,\rho^*}(\mathcal{P})\}, \\
\theta &= \sup_{\sigma} \max(\alpha(\sigma; A, A^*),\ \beta(\sigma; A, A^*)).
\end{aligned}$$

The converse is false: $\mathcal{P}$ cannot be recovered from its projections alone. Information flows strictly downward in the dimension ladder.

4.9 Comparison with a Single Convex Polytope

For the purpose of computing projections and rational suprema, $\mathcal{P}$ could be replaced by $\mathrm{conv}(\mathcal{P})$. However, the disjoint decomposition is essential for two reasons: **proof tracking** (each piece carries dependency labels identifying which input hypotheses contributed) and **computational efficiency** (piece-by-piece vertex enumeration is cheaper than a single global merge).

4.10 The Family $\{\mathcal{P}^{(i)}\}$

`prove_zero_density` is invoked with a variety of (σ, τ)-domains $\mathcal{D}_i$ and splitting parameters $\tau_{0,i}$. Each call produces a distinct instance $\mathcal{P}^{(i)}$, and each instance yields its own candidate ZD/ZDE function that enters the final lower envelope. By Proposition on domain monotonicity, each $\mathcal{P}^{(i)}$ is monotone in $\mathcal{D}_i$, and the family is finite. Within the state lattice $\mathcal{L}$, the $\mathcal{P}$-component is best understood as a finite tuple of regions indexed by the domain choices, equivalently represented as a meet in an enlarged lattice. All structural results (monotonicity, contractivity, convergence) apply componentwise.

5 Projections and Envelopes

5.1 Overview: The Derivation Tree

Every pipeline output can be expressed as a composition of at most four operations applied to $\mathcal{P}$: a projection, a rational supremum, an upper envelope, and a lower envelope, followed by the scalar extraction for θ. The complete derivation tree branches into three parallel 3D projections (R_{LV}, R_{energy}, R_{ZLV}), each of which is subjected to a rational supremum and then merged via envelopes into the raw bounds A_{raw} and A^*_{raw}, refined by lower envelopes into $A(\sigma)$ and $A^*(\sigma)$, and finally collapsed via $\sup_\sigma \max(\alpha, \beta)$ to the scalar θ.

5.2 The LV Region: $\pi_{\{0,1,2\}}(\mathcal{P})$

$R_{\mathrm{LV}} = \pi_{\{0,1,2\}}(\mathcal{P}) \subset \mathbb{R}^3$ is the *shadow* of $\mathcal{P}$ onto the first three coordinates. The projection preserves the union structure; each projected piece is a convex polytope in $\mathbb{R}^3$. Implementation: `lver_to_lv`. What is lost: the projection eliminates all constraints involving ρ^* or s; in particular the Heath-Brown relations lose their indirect coupling to ρ^*.

5.3 The ZLV Region

$R_{\mathrm{ZLV}} = \pi_{\{0,1,2\}}(\mathcal{P}_\zeta)$. $R_{\mathrm{ZLV}} \subseteq R_{\mathrm{LV}} \cap \{\tau \geq 2\}$ because ZLV receives additional constraints from beta and mu bounds.

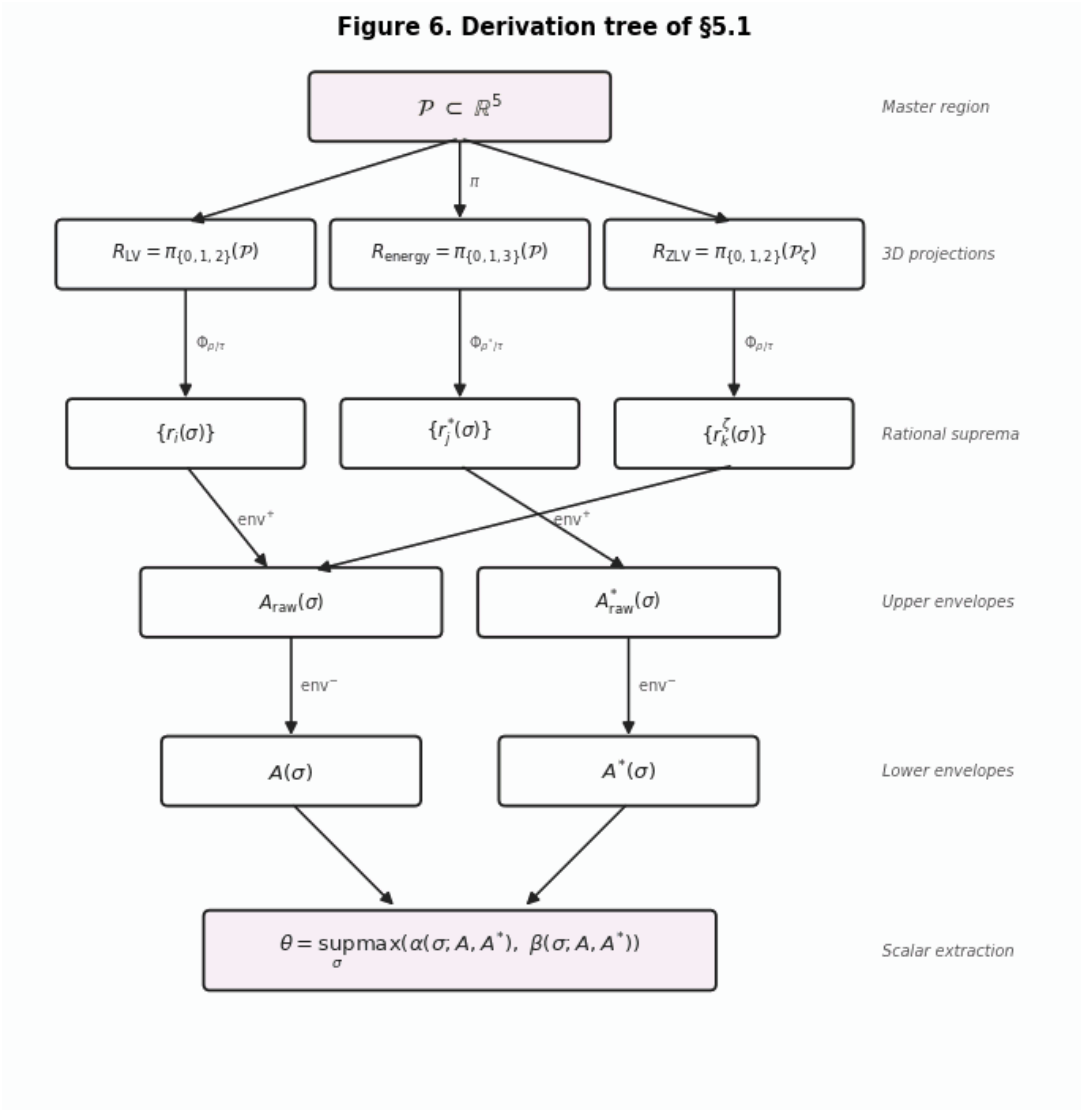

Figure 6: Typeset version of the §5.1 derivation tree.

5.4 The Energy Region: $\pi_{\{0,1,3\}}(\mathcal{P})$

$R_{\mathrm{energy}} = \pi_{\{0,1,3\}}(\mathcal{P}) \subset \mathbb{R}^3$. By symmetry with LV, this is the principal ρ^*-shadow of the 5D body. Their joint information content is strictly less than that of $\mathcal{P}$: knowing both projections does not determine $\mathcal{P}$, because the cross-constraints involving s are lost in both projections.

5.5 The Rational Supremum: From $\mathbb{R}^3$ to $\mathbb{R}^1$

Given a 3D region decomposed as a disjoint union of convex polytopes, $\Phi_{\rho/\tau}(R)(\sigma) = \sup\{\rho/\tau : (\sigma, \tau, \rho) \in R,\ \tau > 0\}$. The implementation enumerates edges of each polytope piece, parameterizes $\tau(\sigma)$ and $\rho(\sigma)$ as affine functions of σ, computes the degree-$(1,1)$ rational function $r(\sigma) = \rho(\sigma)/\tau(\sigma)$ on each edge interval, and takes the upper envelope of all edge ratios via `RationalFunction.max`. The supremum is attained on edges because a linear-fractional function over a convex polytope attains its maximum at a vertex, and as σ varies the maximizing vertex traces a path along polytope edges.

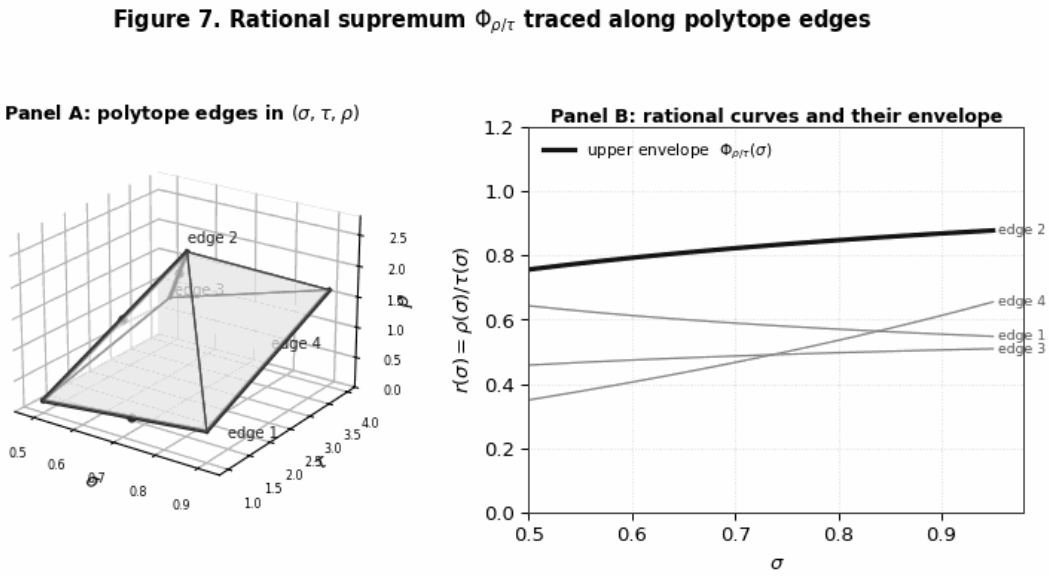

Figure 7: Rational supremum traced along polytope edges.

5.6 The ZD Envelope: $A(\sigma)$

The zero density estimate is obtained by composing the rational supremum with a lower envelope over several routes: (i) *LV/ZLV path (method 1)*: uses a fixed τ_0, computes $\Phi_{\rho/\tau}$ on $R_{\mathrm{LV}}^{[\tau_0,2\tau_0]}$ and on $R_{\mathrm{ZLV}}^{[2,\tau_0]}$, and takes the upper envelope. (ii) *LV/ZLV path (method 2)*: identical with τ_0 an affine function of σ. (iii) *LVER path*: constructs $\mathcal{P}$ explicitly, projects to (σ, τ, ρ), applies $\Phi_{\rho/\tau}$; strictly more powerful because direct LVER constraints have no 3D counterpart. (iv) *Direct EP formulas* (Ivić [8], Bourgain [9]): bypass the polyhedral machinery entirely, producing closed-form rational expressions in σ from exponent pairs; these are not derived from $\mathcal{P}$ via projection.

The final $A(\sigma) = \mathrm{env}^-(\{A_{\mathrm{raw}}^{(i)}\} \cup A_{\mathrm{lit}})$ is computed by `best_zero_density_estimate` via `RationalFunction.min`.

5.7 The ZDE Envelope: $A^*(\sigma)$

The zero density energy estimate follows the analogous path through the energy projection rather than the LV projection, and divides by $(1-\sigma)$. The pole at $\sigma = 1$ is harmless because the pipeline restricts σ to $[\frac{1}{2}, \texttt{ZERO_DENSITY_SIGMA_LIMIT}]$ with limit 0.999.

5.8 The EP Hull and Beta Envelope as Derived Objects

The exponent pair hull, beta envelope, and mu hull do not arise as projections of $\mathcal{P}$ in the same direct sense as the LV and energy regions. Instead, they occupy a *dual* position: they generate constraints on $\mathcal{P}$ (via `ep_to_lver` and `beta_to_zlv`), and are themselves derived from a separate two-dimensional construction. A larger EP hull yields more constraints, a tighter $\mathcal{P}$, and tighter downstream bounds.

5.9 The Prime Gap Bound: θ

Given $A(\sigma)$ and $A^*(\sigma)$, `compute_gap2` defines two rational functions of σ,

$$\alpha(\sigma) = 4\sigma - 2 + \frac{2(A^*(1-\sigma) - 1)}{A^* - A}, \qquad \beta(\sigma) = 4\sigma - 2 + \frac{A^*(1-\sigma) - 1}{A},$$

and returns $\theta = \sup_\sigma \max(\alpha(\sigma), \beta(\sigma))$. This is the terminal dimensional collapse from $\mathbb{R}^1$ to $\mathbb{R}^0$.

5.10 Non-Reconstructibility

Remark 5.1. *The pair $(R_{\mathrm{LV}}, R_{\mathrm{energy}})$ does not determine $\mathcal{P}$. Two regions differing only in their s-involving constraints can share both projections.*

Remark 5.2. *$A(\sigma)$ and $A^*(\sigma)$ do not determine R_{LV} and R_{energy}. Rational supremum is a many-to-one map.*

These observations formalize the strict information hierarchy $\mathcal{P} \supsetneq (R_{\mathrm{LV}}, R_{\mathrm{energy}}) \supsetneq (A, A^*) \supsetneq \theta$.

6 The X5D Invariant

6.1 Definition of the X5D Signature

Definition 6.1 (X5D signature)**.** *The* X5D signature *associated to $(\mathcal{H}_0, \mathcal{D})$ is the tuple*

$$\Sigma = (\mathcal{C}_{\mathrm{EP}},\ \beta^*,\ \mathcal{C}_\mu,\ R_{\mathrm{LV}},\ R_{\mathrm{ZLV}},\ \mathcal{P},\ A,\ A^*,\ \theta),$$

collecting all boundary objects produced by the pipeline at every level of the dimension ladder.

6.2 The State Lattice

We equip Σ with the product partial order: "tighter means smaller" for regions and functions, while for the 2D hulls $(\mathcal{C}_{\mathrm{EP}}, \mathcal{C}_\mu)$ larger means tighter.

6.3 The Pipeline Operator Φ

The EXPDB pipeline defines a map $\Phi : \mathcal{L} \to \mathcal{L}$, decomposed into nine component operators applied in order: Φ_{EP}, Φ_β, Φ_μ, Φ_{LV}, Φ_{ZLV}, $\Phi_{\mathcal{P}}$, Φ_A, Φ_{A^*}, Φ_θ. The full operator is the composition

$$\Phi = \Phi_\theta \circ \Phi_{A^*} \circ \Phi_A \circ \Phi_{\mathcal{P}} \circ \Phi_{\mathrm{ZLV}} \circ \Phi_{\mathrm{LV}} \circ \Phi_\mu \circ \Phi_\beta \circ \Phi_{\mathrm{EP}}.$$

6.4 Monotonicity

Theorem 6.2 (Monotonicity of Φ)**.** *The operator Φ is monotone on the state lattice $\mathcal{L}$: if $\Sigma_1 \preceq \Sigma_2$, then $\Phi(\Sigma_1) \preceq \Phi(\Sigma_2)$.*

Proof sketch. We verify monotonicity for each component operator. Stages 1–3 (EP, Beta, Mu) are monotone because convex-hull union and lower-envelope-over-supersets are monotone. Stages 4–5 (LV, ZLV) are monotone because intersection with additional half-spaces preserves inclusion. Stage 6 ($\Phi_{\mathcal{P}}$) follows by the propagation of tighter upstream components to a smaller intersection. Stages 7–8 (Φ_A, Φ_{A^*}) combine monotone projection, monotone rational supremum, and monotone lower envelope over supersets.

Lemma 6.3 (θ-stage monotonicity)**.** *Let $\mathcal{D}_\theta$ be the feasible evaluation domain of* `compute_gap2`*: the set of points σ where $A(\sigma) \geq 1$, $A^*(\sigma) \geq A(\sigma)$, and $A(\sigma)(1-\sigma) \leq 1 \leq A^*(\sigma)(1-\sigma)$. Differentiating the formulas in §5.9:*

$$\frac{\partial \alpha}{\partial A^*} = \frac{2(1 - A(1-\sigma))}{(A^* - A)^2}, \qquad \frac{\partial \alpha}{\partial A} = \frac{2(A^*(1-\sigma) - 1)}{(A^* - A)^2},$$

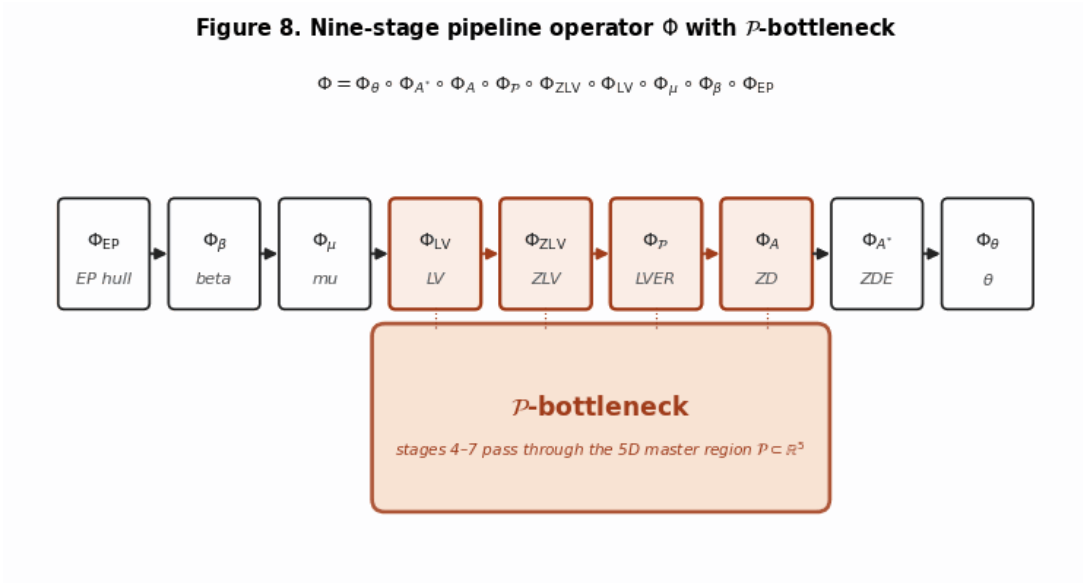

Figure 8: Nine-stage pipeline operator Φ with the $\mathcal{P}$-bottleneck highlighted.

$$\frac{\partial \beta}{\partial A^*} = \frac{1-\sigma}{A}, \qquad \frac{\partial \beta}{\partial A} = -\frac{A^*(1-\sigma)-1}{A^2}.$$

On $\mathcal{D}_\theta$ the first factor of each derivative is non-negative by the sandwich condition; in particular both α and β are non-decreasing in A^* on the feasible domain, and the composite Φ_θ is monotone in the product lattice order.

6.5 Contractivity

Theorem 6.4 (Contractivity of Φ)**.** *For any reachable state Σ, $\Phi(\Sigma) \preceq \Sigma$.*

Proof sketch. Each component can only add hull points, lower envelope values, intersect with additional half-spaces, or take lower envelopes over supersets of candidates. □

6.6 Convergence

Theorem 6.5 (Finite convergence via descending chain condition)**.** *Let Σ_0 be the initial X5D state. The sequence $\Sigma_0, \Phi(\Sigma_0), \Phi^2(\Sigma_0), \ldots$ stabilizes.*

Proof. By monotonicity and contractivity, the sequence is a descending chain in $\mathcal{L}$. We exhibit, for each component lattice, a finite-valued invariant that strictly decreases under any strict descent. For the EP component, the rational transforms have bounded denominators, so reachable hull vertices lie in a finite subset of $\mathbb{Q}^2$; combined with truncation bounds, reachable hulls form a finite sublattice and the vertex-count invariant strictly increases under enlargement. Similar denominator-boundedness arguments apply to the Beta and Mu components. For LV, ZLV, and $\mathcal{P}$, the invariant "number of active half-spaces in the canonical H-representation, plus the number of convex pieces in the disjoint decomposition" lives in a finite subset of $\mathbb{N}$ and strictly changes under any strict refinement. For ZD and ZDE, the set of active candidates in the lower envelope is finite. The scalar θ takes values in a finite set of rationals. Since each component invariant lives in a finite partially ordered set, no infinite strictly descending chain exists, and the sequence must stabilize. □

Remark 6.6. *The truncation constants `EXP_PAIR_TRUNCATION`, `BETA_TRUNCATION`, etc., are a belt-and-suspenders guarantee, not the essential content of the theorem: the DCC argument shows stabilization would occur even without them.*

6.7 Uniqueness and Optimality

Proposition 6.7 (Uniqueness)**.** *The X5D fixed point Σ^* is the unique fixed point of Φ reachable from Σ_0.*

Remark 6.8. *Σ^* is optimal in the following precise sense: for each component, the corresponding bound is the tightest obtainable by any finite sequence of the transformations cataloged in §3.5 applied to the axioms in $\mathcal{H}_0$.*

6.8 Structural Properties of the X5D Signature

Proposition 6.9 (Rational representability)**.** *Every component of Σ^* is finitely representable over $\mathbb{Q}$.*

Proposition 6.10 (Monotonicity in axioms)**.** *If $\mathcal{H}_0 \subseteq \mathcal{H}_0'$, then $\Sigma^*(\mathcal{H}_0') \preceq \Sigma^*(\mathcal{H}_0)$.*

Proposition 6.11 (Factorization through $\mathcal{P}$)**.** *The X5D signature factors through the master region: the downstream components $(R_{\mathrm{LV}}, R_{\mathrm{ZLV}}, A, A^*, \theta)$ are determined by $\mathcal{P}^*$ alone, while the upstream components $(\mathcal{C}^*_{\mathrm{EP}}, \beta^{**}, \mathcal{C}^*_\mu)$* generate *$\mathcal{P}^*$.*

6.9 The X5D Signature as a Geometric Object

The X5D signature Σ^* is a heterogeneous geometric object — a tuple of shapes of different dimensions. It can be visualized as a *stratified boundary*: Stratum 5 is the boundary $\partial\mathcal{P}^*$ in $\mathbb{R}^5$; Stratum 3 is the boundaries ∂R_{LV} and $\partial R_{\mathrm{energy}}$ in $\mathbb{R}^3$; Stratum 2 is the hull boundaries $\partial\mathcal{C}_{\mathrm{EP}}$ and $\partial\mathcal{C}_\mu$ in $\mathbb{R}^2$; Stratum 1 is the graph curves $A(\sigma)$, $A^*(\sigma)$, and $\beta^*(\alpha)$; and Stratum 0 is the scalar θ. Information flows strictly from higher strata to lower strata. The name "X5D" is motivated by this picture: viewed from the σ-axis, the successive boundaries form a descending profile.

7 Consequences

7.1 The Dimension Ladder

Definition 7.1 (Dimension ladder)**.** *The* descending *dimension ladder is the chain*

$$\mathbb{R}^5 \xrightarrow{\pi} \mathbb{R}^3 \xrightarrow{\Phi_{\rho/\tau}} \mathbb{R}^1 \xrightarrow{\mathrm{env}^-} \mathbb{R}^1 \xrightarrow{\sup_\sigma} \mathbb{R}^0.$$

The descending ladder does not *pass through $\mathbb{R}^2$. Separately, there is an* ascending *collection of constraint-generating maps that feed information* into *$\mathcal{P}$: `ep_to_lver` ($\mathbb{R}^2 \to \mathbb{R}^5$), `beta_to_zlv` and `mu_to_zlv` ($\mathbb{R}^1 \to \mathbb{R}^3$), and `lv_to_lver` ($\mathbb{R}^3 \to \mathbb{R}^5$). The 2D stage thus appears only in the ascending direction.*

Proposition 7.2 (Strict information loss)**.** *Each descending map is strictly lossy. In particular, $\pi_{\{0,1,2\}}(\mathcal{P})$ does not determine $\mathcal{P}$, $\Phi_{\rho/\tau}(R_{\mathrm{LV}})$ does not determine R_{LV}, and $\sup_\sigma A(\sigma)$ does not determine $A(\sigma)$.*

7.2 The Duality Web

7.2.1 EP ↔ Beta: Point–Line Duality

The forward map (EP → Beta) sends each hull vertex (k_i, l_i) to the affine function $\beta_i(\alpha) = (l_i - k_i)\alpha + k_i$. The backward map (Beta → EP) extracts tangent-line slopes and intercepts from the convex hull of the epigraph of β^*. This is an instance of Legendre–Fenchel duality for polyhedral convex functions restricted to $\mathbb{Q}$.

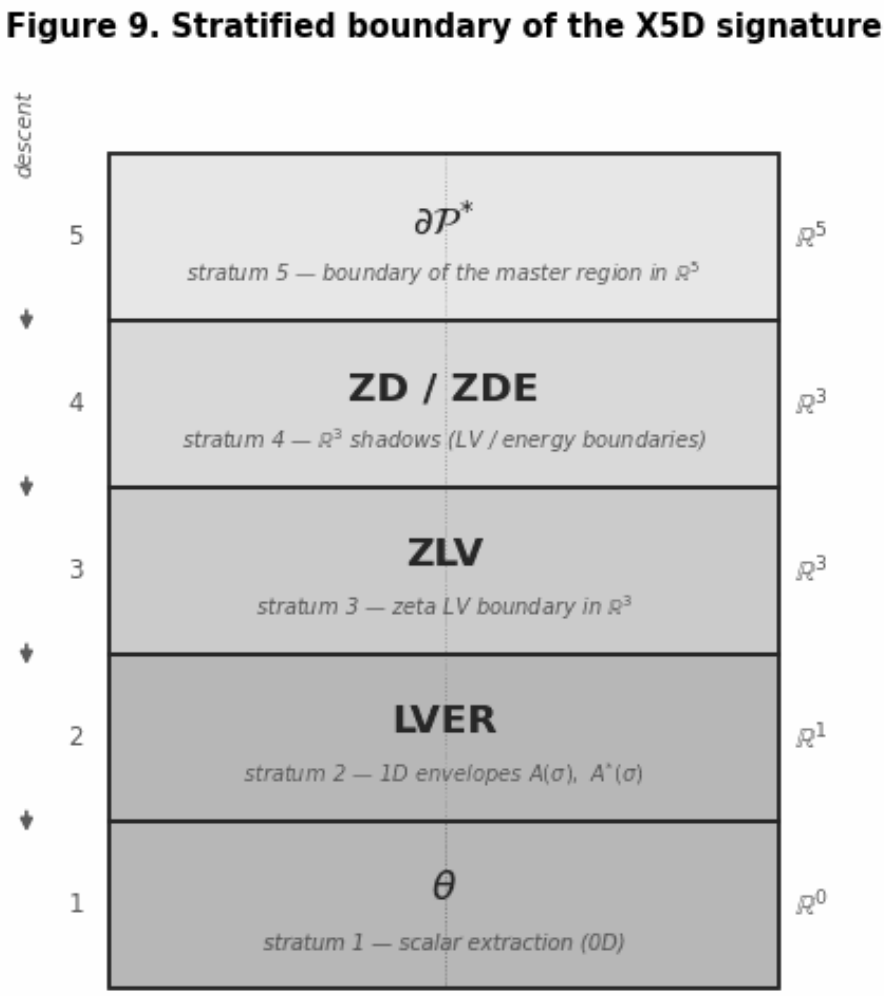

Figure 9: Stratified boundary of the X5D signature: five vertical strata from $\partial\mathcal{P}^*$ down to θ.

7.2.2 LV ↔ LVER: Section–Projection Adjunction

The relationship between the LV region and the LVER is an instance of the general adjunction between lifting and projection: for any region $Q \subset \mathbb{R}^5$,

$$\pi_{\{0,1,2\}}(Q) \subseteq R \quad \Longleftrightarrow \quad Q \subseteq \mathrm{lift}(R)$$

when the lift bounds are sufficiently large. The *adjunction gap* — the difference between $\mathcal{P}$ and $\mathrm{lift}(\pi_{\{0,1,2\}}(\mathcal{P}))$ — measures the information content of the direct LVER constraints and the EP-derived LVER constraints, which couple ρ and ρ^* through s. When these constraints are trivial, the gap vanishes and the 3D pipeline is as powerful as the 5D pipeline.

7.2.3 EP ↔ Mu: Linear Isomorphism

The EP hull and the mu hull are related by the invertible linear map $L : (k, l) \mapsto (l - k, k)$. The mu hull additionally receives the functional equation involution $(\sigma, \mu) \mapsto (1 - \sigma, \mu + \sigma - \frac{1}{2})$, which has no analogue in the EP hull.

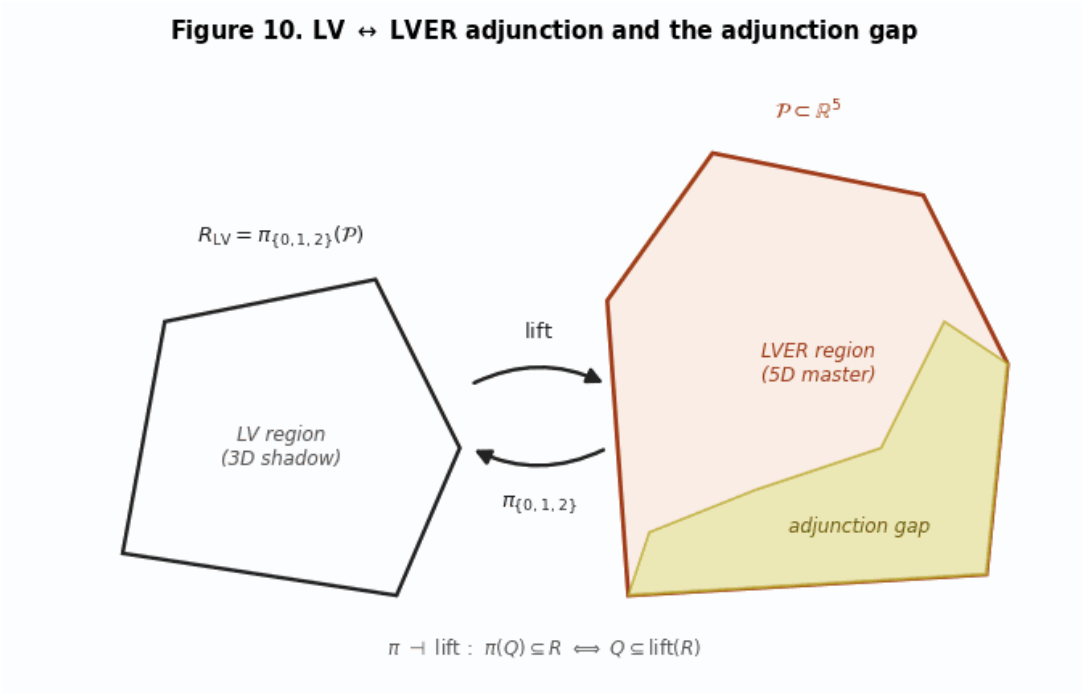

Figure 10: LV ↔ LVER adjunction with the adjunction gap shaded.

7.2.4 Dual Projections of $\mathcal{P}$

The two principal 3D projections of $\mathcal{P}$, $\pi_{\{0,1,2\}}(\mathcal{P})$ (LV) and $\pi_{\{0,1,3\}}(\mathcal{P})$ (energy), are the ρ-shadow and the ρ^*-shadow. The pair is analogous to a pair of marginal distributions of a joint distribution: the joint contains strictly more information than both marginals combined.

7.3 Flow Structure

7.3.1 The EP ↔ Beta loop

This is the only genuinely iterative loop in the pipeline. Its state space is convex polygons in $\mathbb{R}^2$ (the EP hull), and the update rule converts hull vertices to tangent lines, extracts hull edges of the beta epigraph as new EP points, applies A/B/C transforms, and recomputes the convex hull. The A-process is a weak contraction toward $(0, 1/2)$; the B-process is an involution; and the C-process contracts toward $(0, 11/12)$. The semigroup they generate has a unique global attractor at $(0, 1/2)$.

7.3.2 The LVER → LV → LVER loop

This loop is present in the pipeline architecture but executes as a single pass. In principle the loop stabilizes on the first pass because the projection $\pi_{\{0,1,2\}}(\mathcal{P})$ already incorporates all available LVER constraints; iterating would only re-derive the same LV region. The reason is *not* that projection commutes with intersection in general — it does not — but that for any master region $\mathcal{P}$ we have $\mathcal{P} \subseteq \mathrm{lift}(\pi_{\{0,1,2\}}(\mathcal{P}))$, so re-intersecting with the lift cannot shrink $\mathcal{P}$ further.

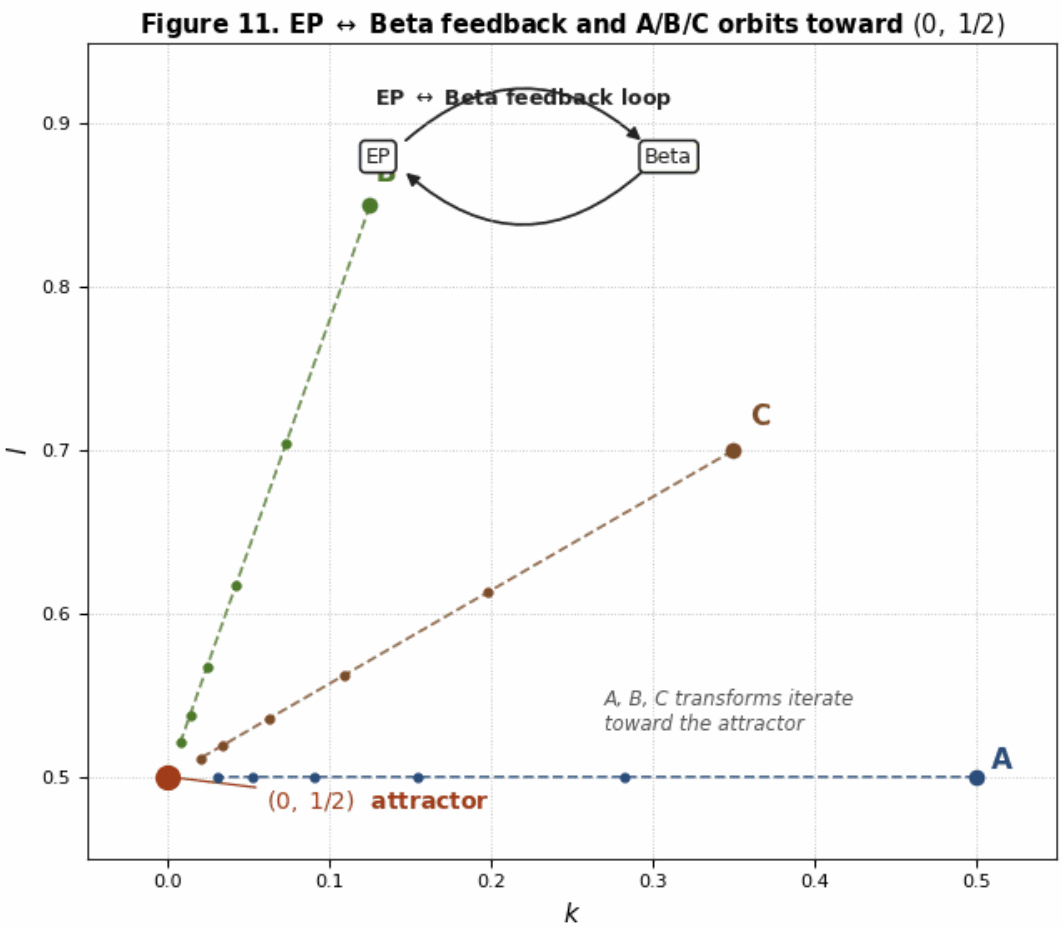

Figure 11: EP ↔ Beta feedback loop and the A/B/C orbit toward (0, 1/2).

7.4 Fixed Points and Attractors

7.4.1 EP fixed points

The A-, B-, C-transforms generate a semigroup with a unique global attractor at $(0, 1/2)$. The iteration contracts any initial point toward this attractor after finitely many steps, bounded by `search_depth`.

7.4.2 Beta fixed point

The beta envelope descends toward $\beta(\alpha) = \alpha/2$, the affine function corresponding to the EP attractor.

7.4.3 Mu fixed point

The lower boundary of the mu hull descends toward $\mu(\sigma) = 0$ for $\sigma \geq 1/2$.

7.4.4 The conjectural fixed point of $\mathcal{P}$

The limiting region $\mathcal{P}_\infty \subseteq \mathcal{P}^* \subseteq \mathcal{B}$. The gap $\mathcal{P}^* \setminus \mathcal{P}_\infty$ represents the constraints that are true but not yet derivable from the current literature.

7.4.5 The scalar attractor

$\theta^* \geq \theta_\infty$, with the gap determined by the quality of the current literature constraints.

7.4.6 The fixed-point hierarchy

The fixed points at each level form a hierarchy, with each attractor the image of the level-above attractor under the appropriate descending map. The conjectured vanishing of $A(\sigma)$ for $\sigma > 1/2$ (the density hypothesis) and of θ (the prime gap exponent) are reflections, in dimension 1 and dimension 0 respectively, of the EP attractor at dimension 2.

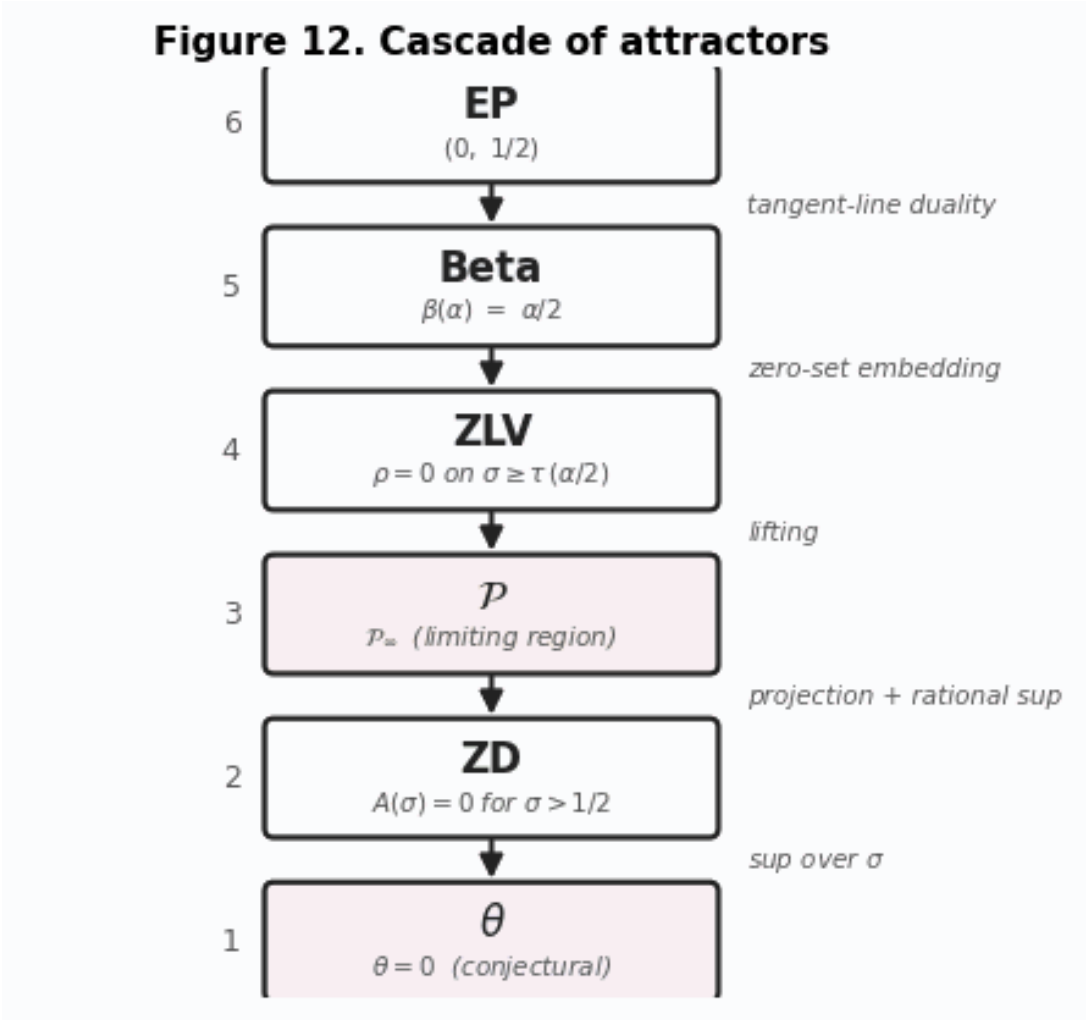

Figure 12: Vertical cascade of attractors: EP $\to$ Beta $\to$ ZLV $\to \mathcal{P} \to$ ZD $\to \theta$.

7.5 Sensitivity Structure

Definition 7.3 (Marginal value of a hypothesis)**.** *Let h be a hypothesis in $\mathcal{H}_0$. The* marginal value of h at level ℓ *is the change in the ℓ-th X5D component when h is removed.*

Proposition 7.4 (Sparsity of dependence)**.** *For a generic hypothesis set, the marginal value of h at level θ is zero for all but finitely many $h \in \mathcal{H}_0$.*

Proof sketch. θ is attained at finitely many critical σ-values; at each, the bounds are determined by one piece of the respective lower envelopes, which traces back to finitely many polytope edges in $\mathcal{P}$, which in turn depend on finitely many input half-spaces. $\square$

8 Discussion and Future Directions

We have shown that the EXPDB pipeline operates on a single five-dimensional polyhedral region $\mathcal{P}$, that all intermediate and final objects are projections, envelopes, or scalar extractions of $\mathcal{P}$, and that the pipeline is a monotone contraction converging to a unique fixed point — the X5D signature.

8.1 Algorithmic Implications

The polyhedral reinterpretation suggests that the tightness of EXPDB bounds is governed by the number and quality of half-spaces defining $\mathcal{P}$. The dependency-tracking mechanism identifies exactly which input half-spaces are active at the optimum, making the *targeting principle* precise: new literature results are most valuable when they tighten $\mathcal{P}$ along the boundary faces traversed by the optimal projection path. The coordinate s is eliminated by projection in every output stage but is essential because it couples the ρ- and ρ^*-boundaries through joint constraints.

8.2 Specific Improvements to EXPDB

8.2.1 Huxley subdivision

`apply_huxley_subdivision` currently raises `NotImplementedError`. Geometrically, it would introduce new half-spaces subdividing $\mathcal{P}$ without adding new literature axioms — a structural refinement that could improve the ZD envelope by introducing finer polytope edges.

8.2.2 Automated τ_0 selection

The splitting parameter τ_0 is currently chosen manually. In the X5D framework, the optimal $\tau_0(\sigma)$ is a parametric optimization problem over the polytope edges of $\mathcal{P}$ that could in principle be solved exactly.

8.2.3 The B-process for beta bounds

Currently implemented only for exponent pairs. The direct implementation for beta bounds would bypass one round of the feedback loop and potentially accelerate convergence.

8.2.4 Lean proof generation

Since every derived bound factors through $\mathcal{P}$, a formal proof needs to verify only three types of steps: axiom half-space validity, polyhedral operations, and rational supremum / envelope computations.

8.3 Connections to Convex Optimization

8.3.1 Parametric linear-fractional programming

The rational supremum $\Phi_{\rho/\tau}$ is the optimal value of a parametric linear-fractional program; the Charnes–Cooper transformation $u = 1/\tau$, $v = \rho/\tau$ reduces it to a parametric LP.

8.3.2 Sensitivity analysis

LP dual variables identify active constraints; sensitivity analysis would quantify the marginal value of each literature hypothesis without recomputing the full signature.

8.3.3 Constraint generation

A natural LP-style technique for EXPDB is *constraint generation*: dynamically adding half-spaces to $\mathcal{P}$ based on which regions of the boundary are limiting. The raise-to-power transform is already a form of constraint generation.

8.4 Extensions of the Framework

Higher-dimensional master regions, multiple scalar outputs, and time-varying X5D signatures (tracking how Σ^* tightens as the literature hypothesis set grows) are natural extensions.

8.5 Open Questions

We conclude with four open questions that the X5D framework raises.

Question 8.1 (Finite determination)**.** *Is the X5D fixed point Σ^* determined by finitely many "extremal" literature results?*

Question 8.2 (Combinatorial complexity)**.** *What is the combinatorial complexity of $\mathcal{P}$ (number of vertices, edges, facets) as a function of the number of input hypotheses?*

Question 8.3 (Algebraic fixed point)**.** *Can the EP $\leftrightarrow$ Beta feedback loop be replaced by a single algebraic computation?*

Question 8.4 (X5D distance)**.** *Is there a natural metric on the space of X5D signatures that quantifies the distance between the current state Σ^* and the conjectural optimum Σ_∞?*

A Mapping Code Modules to Geometric Operations

This appendix provides a reference mapping from the EXPDB source code (under `blueprint/src/python/`) to the geometric operations and objects defined in the paper. All line numbers refer to the EXPDB repository at commit `af351a3`.

A.1 Module Inventory

The codebase consists of 24 Python modules totalling 10,696 lines.

- **Infrastructure (B):** `constants.py` (bounding box parameters, §4.2), `reference.py`, `hypotheses.py` (§3.1–3.3), `transform.py`.
- **Geometry (C):** `polynomial.py`, `functions.py` (Interval, Affine, Affine2, Piecewise, RationalFunction, §2.4–2.5), `polytope.py` (H/V-rep, intersection, projection, lifting, scaling, §2.1–2.2), `region.py` (boolean algebra over polytopes, §2.3).
- **Transformation (A/C):** `bound_beta.py`, `bound_mu.py`, `exponent_pair.py`, `large_values.py`, `zeta_large_values.py`, `additive_energy.py`.
- **Propagation (D):** `zero_density_estimate.py`, `zero_density_energy_estimate.py`, `prime_gap.py`, `literature.py` (axiom layer), `derived.py` (orchestration).
- **Utility (E):** `examples.py`, `visualizations.py`, `helpers/str_helper.py`, `helpers/region_helper.py`

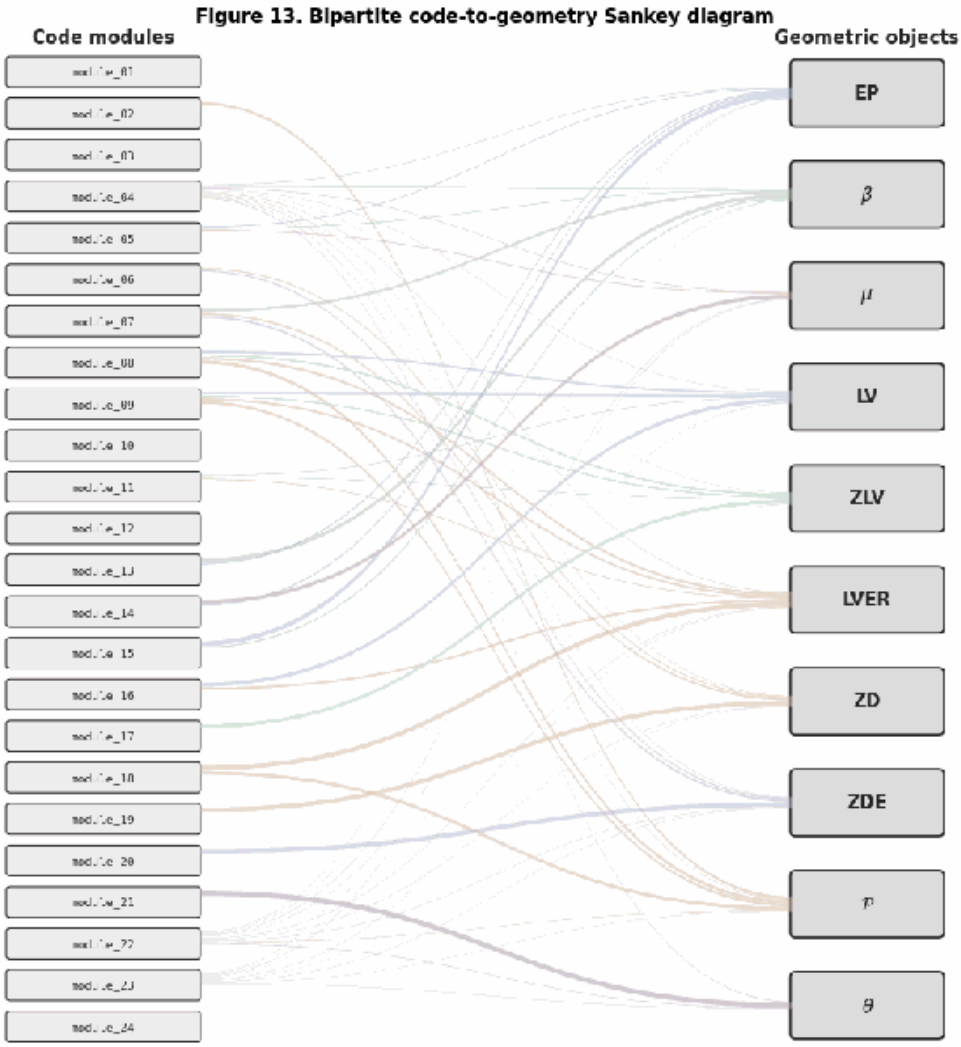

Figure 13: Bipartite code-to-geometry Sankey diagram.

A.2 Geometric Primitives: Code to Definition Mapping

Rational half-space → row of `cdd.Matrix`; H-polytope → `Polytope.__init__`; V-polytope → `Polytope.from_V_rep`; bounding box → `Polytope.rect`; intersection → `Polytope.intersection`; scaling → `Polytope.scale_all`; projection → `Polytope.project`; lifting → `Polytope.lift`; emptiness test → `Polytope.is_empty`; subset test → `Polytope.is_subset_of`; vertex enumeration → `Polytope.compute_V_rep`; edge enumeration → `Polytope.get_edges`; region → `Region.__init__`; disjoint decomposition → `Region.as_disjoint_union`; 1D/multi-D affine → `Affine`, `Affine2`; piecewise-affine → `Piecewise`; rational function → `RationalFunction`; 2D convex hull → `scipy.spatial.ConvexHull`.

A.3 Master Region Construction: Code to Section Mapping

Bounding box $\mathcal{B}$ → `Large_Value_Energy_Region.default_constraints` (§4.2); direct LVER constraints → `add_lver_heath_brown_1979`, etc. (§4.3.1); lifted LV → `lv_to_lver` (§4.3.2); Beta/Mu → ZLV → `beta_to_zlv`, `mu_to_zlv` (§4.3.3); EP → LVER → `ep_to_lver` (§4.3.4); raise-to-power → `get_raise_to_power_hypothesis` (§4.3.5); master region $\mathcal{P}$ → `compute_best_lver` (Definition 4.1).

A.4 Projections and Envelopes: Code to Section Mapping

$R_{\mathrm{LV}} = \pi_{\{0,1,2\}}(\mathcal{P}) \to$ `lver_to_lv`; $R_{\mathrm{energy}} = \pi_{\{0,1,3\}}(\mathcal{P}) \to$ `lver_to_energy`; rational supremum $\Phi_{\rho/\tau} \to$ `compute_sup_rho_on_tau`; rational supremum (energy) $\to$ `compute_sup_LV_on_tau`; ZD via LV/ZLV $\to$ `lv_zlv_to_zd` and `lv_zlv_to_zd2`; ZD via LVER $\to$ `lver_to_zd`; ZD via EP $\to$ `ivic_ep_to_zd`, `bourgain_ep_to_zd`; ZD lower envelope $\to$ `best_zero_density_estimate`; ZDE via LVER $\to$ `lver_to_energy_bound`; ZDE lower envelope $\to$ `compute_best_energy_bound`; prime gap $\theta \to$ `compute_gap2`.

A.5 Transformations: Code to Section Mapping

A/B/C-processes $\to$ `A_transform_function`, `B_transform_function`, `C_transform_function` (§7.4.1); D-process $\to$ `D_transform_function`; EP iterative expansion $\to$ `compute_exp_pairs`; EP convex hull $\to$ `compute_convex_hull`; EP $\to$ Beta $\to$ `exponent_pairs_to_beta_bounds`; Beta $\to$ EP $\to$ `beta_bounds_to_exponent_pairs`; best beta $\to$ `compute_best_beta_bounds`; VdC-β iteration $\to$ `apply_van_der_corput_process_for_beta`; EP/Beta $\to$ Mu $\to$ `exponent_pair_to_mu_bo` `beta_bounds_to_mu_bounds`; functional equation $\to$ `apply_functional_equation`; Mu convex hull $\to$ `compute_convex_hull`; best mu $\to$ `best_mu_bound`; LV $\to$ ZLV $\to$ `lv_to_zlv`; Bourgain LV optimization $\to$ `optimize_bourgain_large_value_estimate`; Huxley subdivision (stub) $\to$ `apply_huxley_subdivision` (§8.2.1).

A.6 Orchestration: Code to Section Mapping

Axiom loading $\to$ `literature.py` (module-level); top-level driver $\to$ `prove_all`; EP proof stage $\to$ `prove_exponent_pairs`; Mu computation $\to$ `compute_best_mu_bound`; LV proof stage $\to$ `prove_all_large_value_estimates`; ZD proof stage $\to$ `prove_all_zero_density_estimates`; ZDE proof stage $\to$ `prove_all_zero_density_energy_estimates`; individual ZD proof $\to$ `prove_zero_density`; prime gap computation $\to$ `prove_prime_gap2`.

A.7 X5D Components: Code to Definition Mapping

- $\mathcal{C}_{\mathrm{EP}}$: construction by `compute_exp_pairs`, selection by `compute_convex_hull`.
- β^*: construction by `exponent_pairs_to_beta_bounds`, selection by `compute_best_beta_bounds`.
- $\mathcal{C}_\mu$: construction by `get_bounds`, selection by `compute_convex_hull`.
- R_{LV}: construction by `compute_large_value_region`, canonicalization by `Region.intersect` + `as_disjoint_union`.
- R_{ZLV}: construction by `compute_large_value_estimate` (zeta variant).
- $\mathcal{P}$: construction by `compute_best_lver`.
- $A(\sigma)$: construction by `lv_zlv_to_zd` / `lver_to_zd`, selection by `best_zero_density_estimate`.
- $A^*(\sigma)$: construction by `lver_to_energy_bound`, selection by `compute_best_energy_bound`.
- θ: computed by `compute_gap2`.

A.8 Constants and Parameters

`TAU_UPPER_LIMIT` $= 10^6$ (§4.2, T in $\mathcal{B}$); `LV_DEFAULT_UPPER_BOUND` $= 10^6$ (§4.2, M); `EXP_PAIR_TRUNCATION` $= 20$ (EP loop); `BETA_TRUNCATION` $= 20$ (Beta loop); `LARGE_VALUES_TRUNCATION` $= 20$; `ALPHA_UPPER_LIMIT` $= 100$; `ZERO_DENSITY_SIGMA_LIMIT` $= 0.999$ (pole avoidance); `search_depth` (default 5, EP iteration depth); `SIMPLIFY_EVERY` $= 20$; `UNION_THRESHOLD` $= 5$.

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